

Expense Tracker - Technical Specification & Architecture Document

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1. Executive Summary & Architecture Overview

1.1 Executive Brief

The Expense Tracker is a specialized data-model implementation designed for recording financial expenditures. The system centers on a single Expense entity with strict validation for monetary amounts and temporal markers, utilizing a simplified state machine restricted to creation and deletion operations. The data pattern follows a flat entity structure with ISO-standardized attributes for currency and dating.

1.2 Maturity Assessment

The specification is currently in a state of REFINEMENT. While the data model is complete for the Expense entity, the project suffers from significant structural voids, specifically the total absence of high-priority business goals, non-functional requirements, and formal scope boundaries, which prevents full implementation readiness.

1.3 Technical Stack

- **ID Generation**: ``Date.now().toString()``
- **Date Standard**: ISO 8601
- **Currency Standard**: ISO Currency Codes

1.4 Architectural Constraints

- **Amount Validation**: Must be a positive number strictly > 0 .
- **Date Validation**: Must adhere to ISO 8601 format and cannot be a future date.
- **State Transition Restriction**: Only creation and deletion are permitted; edits and soft-deletes are explicitly excluded.

1.5 Critical Dependencies

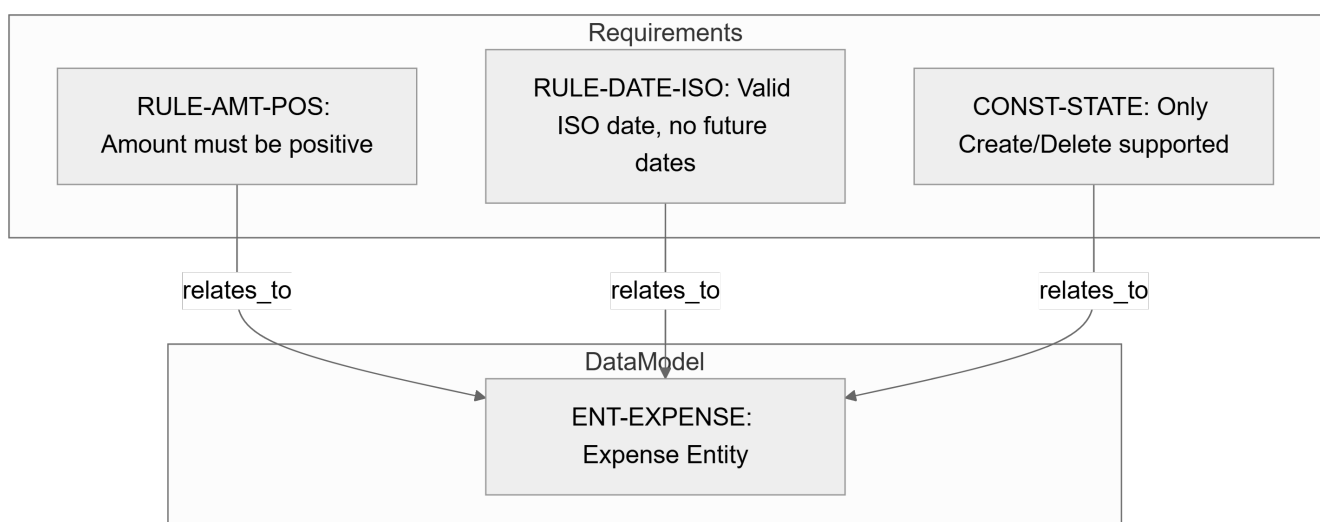
- ISO 8601 date standard compliance.
- ISO currency code standard for the currency attribute.
- Strict referential integrity between validation rules (``RULE-AMT-POS``, ``RULE-DATE-ISO``) and the ``ENT-EXPENSE`` entity.

2. Architecture Workflows & Visual Diagrams

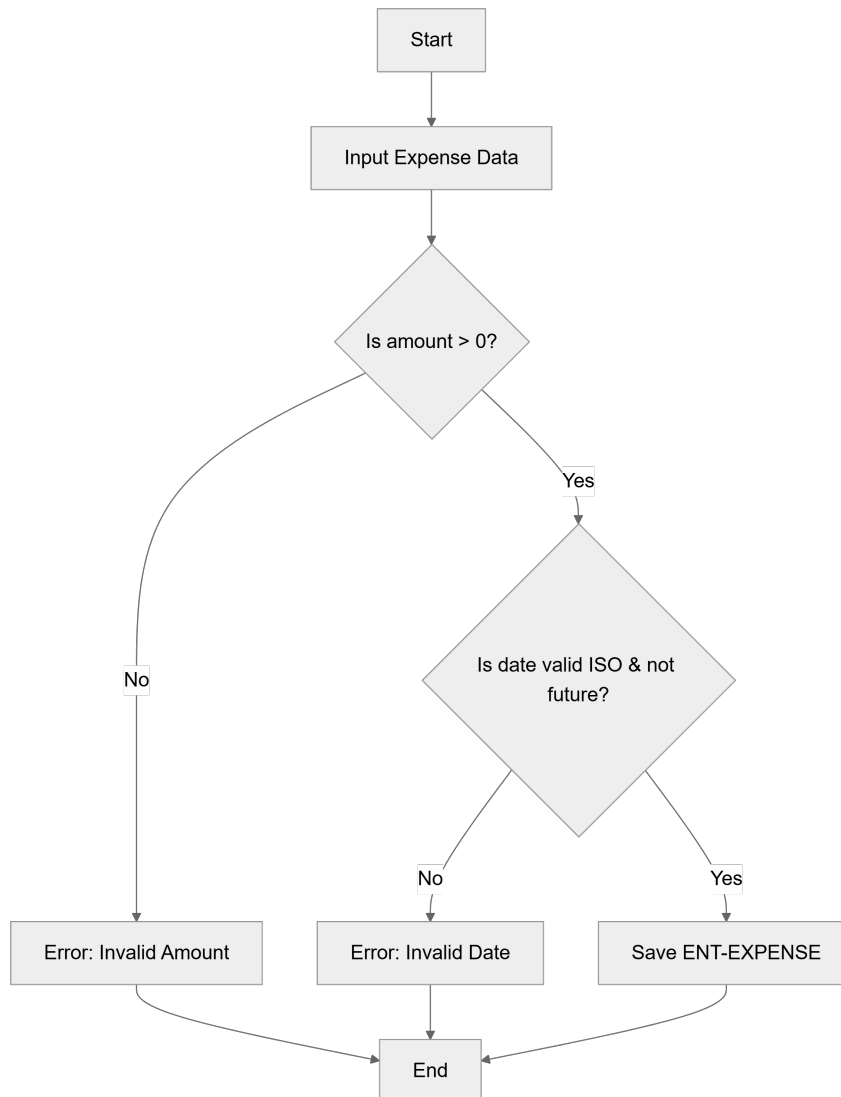
2.1 Expense Data Model

ENT-EXPENSE		
string	id	PK
number	amount	
string	currency	
string	category	
string	description	
string	date	

2.2 Expense Validation & Constraint Traceability



2.3 Expense Creation Workflow



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

Identifier	Type	Description	Source Section
`ENT-EXPENSE`	Entity	Expense entity consisting of id, amount, currency, category, description, and date.	data-model.md
`RULE-AMT-POS`	Functional Requirement	The amount must be a positive number (>0).	data-model.md
`RULE-DATE-ISO`	Functional Requirement	The date must be a valid ISO 8601 date and cannot be in the future.	data-model.md
`CONST-STATE`	Constraint	Only creation and deletion are supported; edits and soft-deletes are explicitly excluded in the initial version.	data-model.md

3.2 Security Rules

No specific security rules were defined in the source data.

3.3 Data Models

Entity: ENT-EXPENSE

Attribute	Type	Constraints
`id`	string	Generated via `Date.now().toString()`
`amount`	number	Required, > 0 (`RULE-AMT-POS`)
`currency`	string	ISO code, default `USD`
`category`	string	Optional
`description`	string	Optional
`date`	string	Required, ISO 8601, not in future (`RULE-DATE-ISO`)

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Missing Section	Priority	Remediation Advice
Goals & Objectives	HIGH	Define the business purpose of the Expense Tracker.
Non-Functional Requirements	MEDIUM	Specify performance, security, or accessibility requirements.
Scope & Out-of-Scope	MEDIUM	Formally define the boundaries of the application.
Open Questions & Uncertainties	LOW	List any unresolved architectural or business decisions.

4.2 Remediation & Workflow

The project must transition from the current "Refinement" state to "Implementation Ready" by addressing the high-priority gaps in business goals and scope definition.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
Fixed-Point Numeric Constraint	TECHNICAL_STACK	`RULE-AMT-POS`	A validation requirement ensuring the monetary value remains strictly greater than zero.
USD	BUSINESS_DOMAIN	`ENT-EXPENSE`	The default three-letter international standard code for the United States Dollar used as the baseline monetary unit.